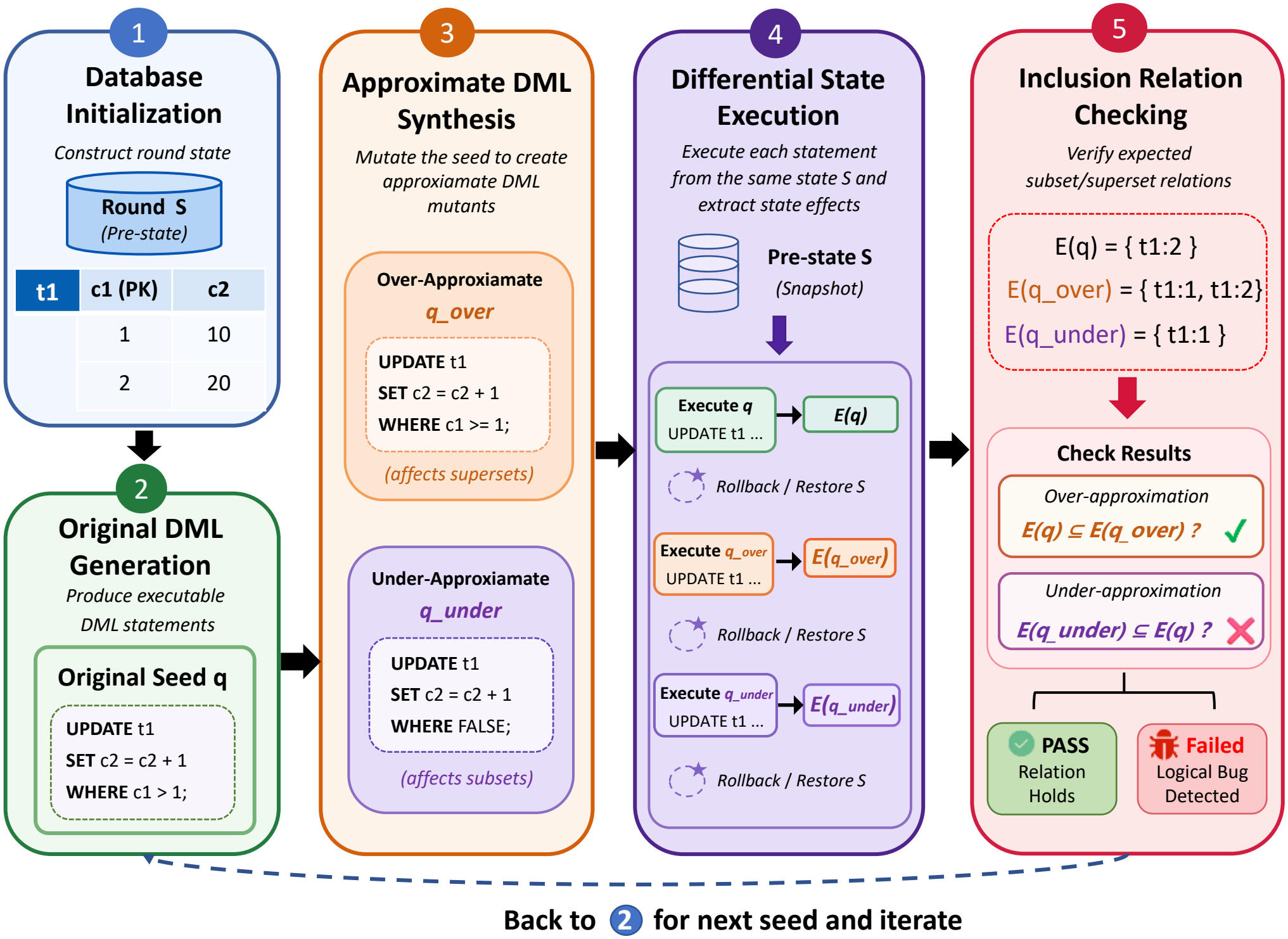
1
Database Initialization
Construct round state
Round S
(Pre-state)
t1
c1 (PK)
c2
1
10
2
20
2
Original DML Generation
Produce executable DML statements
Original Seed q
UPDATE t1
SET c2 = c2 + 1
WHERE c1 > 1;
3
Approximate DML Synthesis
Mutate the seed to create approximate DML mutants
Over-Approximate
q_over
UPDATE t1
SET c2 = c2 + 1
WHERE c1 >= 1;
(affects supersets)
Under-Approximate
q_under
UPDATE t1
SET c2 = c2 + 1
WHERE FALSE;
(affects subsets)
4
Differential State Execution
Execute each statement from the same state S and extract state effects
Pre-state S
(Snapshot)
Execute q
UPDATE t1 ...
E(q)
Rollback / Restore S
Execute q_over
UPDATE t1 ...
E(q_over)
Rollback / Restore S
Execute q_under
UPDATE t1 ...
E(q_under)
Rollback / Restore S
5
Inclusion Relation Checking
Verify expected subset/superset relations
E(q) = { t1:2 }
E(q_over) = { t1:1, t1:2}
E(q_under) = { t1:1 }
Check Results
Over-approximation
E(q) ⊆ E(q_over) ?
Under-approximation
E(q_under) ⊆ E(q) ?
PASS
Relation Holds
Failed
Logical Bug Detected
Back to 2 for next seed and iterate